Data Visualization in R

Workshop Lead: Meghana Munipalle

Approximate duration: 2 hours

Prerequisites:

1. Previous experience with R (basic understanding of syntax, data structures (data frames), and running scripts)
2. Experience with data manipulation using *dplyr* is recommended

Summary:
This workshop will introduce participants to creating flexible, aesthetic and informative visualizations in R for statistical computing through the Grammar of Graphics framework. The seminar explores the fundamental concepts and structure of ggplot2 commands and will guide you in selecting the most effective visualizations for your data. Moving to advanced customization, participants will learn techniques for transforming data with scales, using faceting for complex comparisons, and enhancing visual aesthetics through themes and formatting options.

The seminar will also dive into specialized plots (e.g., heatmaps, Kaplan-Meier curves) and explore interactive features using the plotly package and showcase extending the former through the Shiny web application framework. The workshop will emphasize the importance of working with and understanding your data, to allow for tailoring visualizations to extract the most out of your data for your audience, equipping participants with a versatile toolkit for effective data storytelling.

Learning Objectives:

1. Understand how to use R's *ggplot2* package to create plots.
2. Learn how to select the right type of visualization for your data.
3. Learn how to customize your visualizations for more effective storytelling.
4. Know how to get started with specialized plot formats, animations, interactive plots, and the Shiny web app for interactive dashboards to present your data.

Content:

1. **Module 1: Brief Review of Basic Visualization in R**
    - Why use ggplot 2
    - Structure of a ggplot2 command
2. **Module 2: Advanced Customization and Data Transformation**
    - Data transformations and faceting
    - Customizing ggplot2 plots

- *Exercise 1*

3. **Module 3: Advanced Visualization Techniques**
    - Specialized plots
    - Introduction to interactive and dynamic visualizations
    - *Exercise 2*